Confidence interval for the difference of two means. Independent samples, variance known

University example

	Engineering	Management	Difference
Size	100	70	?
Sample mean	58	65	-7.00
Population std	10	5	1.16

95% CI, $z_{0.025}$ 1.96

Confidence intervals

Z	CI low	CI high
95%	-9.28	-4.72

The formula to find this interval is:

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

To understand this formula, let's break down each part:

- $(\bar{x}_1 - \bar{x}_2)$: This is the difference between the averages of your two groups. It is your "best guess" for what the real difference is.
- t^*: This is called the **critical value**. It depends on how confident you want to be (usually 95%) and the "degrees of freedom" (which comes from the size of your groups).
- s_1 **and** s_2: these are the **standard deviations** of each group. They tell us how "spread out" the data is in each group.
- n_1 **and** n_2: These are the **sample sizes**. It's simply the number of people or items you measured in each group.
- $\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$: This whole part is called the **standard error**. It measures how much we expect our "guess" to jump around if we did the study again.